

Yangoti Sai Bhavya

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Skills

Languages	Python, C, C++, SQL
Frameworks	FastAPI, LangChain, LangGraph, PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, NumPy, Pandas
Tools	Git, GitHub, Docker, Ollama, OpenRouter, Kestra, AWS S3, Streamlit, Gradio, Linux
Databases	MySQL, MongoDB, ChromaDB, Vector Databases
AI & ML	Large Language Models (LLMs), Vision Language Models (VLMs), Prompt Engineering, Fine-Tuning (LoRA, QLoRA, PEFT), Retrieval-Augmented Generation (RAG), Semantic Search, Hybrid Retrieval, Multi-Agent Systems, Tool Calling, Workflow Orchestration, NLP, Computer Vision, Deep Learning, Multimodal AI, Model Evaluation, Inference Optimization

Experience

Timepilot, Remote

Feb 2026 – Present

ML & GenAI Engineer Intern

- Designing and developing multimodal AI pipelines to transform complex financial documents into podcasts, audio briefings, and multimedia summaries for improved accessibility and user engagement.
- Building LangGraph ReAct multi-agent workflows with structured prompting and tool-calling, driving content generation across multiple financial content formats including spotlights, filings, and social posts.
- Integrating LLMs, VLMs, and TTS engines including GPT-4o, Kyutai, and Voxtral through OpenRouter APIs with MongoDB and AWS S3 for end-to-end multimodal AI pipeline execution.
- Parallelized company search and retrieval pipelines, cutting query latency from 10-15s to 3-5s, directly improving real-time responsiveness for financial content workflows.
- Delivered production-grade Dockerized workflows with Kestra-based orchestration and retry logic, reducing manual intervention by ~70% and improving pipeline reliability across automated content execution.
- **Tech Stack:** Python, LangGraph ReAct Agents, Kestra, MongoDB, AWS S3, Docker, OpenRouter, GPT-4o

IITM Pravartak Foundation, Chennai

Apr 2025 – Jul 2025

AI Project Intern

- Developed an end-to-end LLM-powered ETL pipeline for processing 500+ unstructured government audit PDFs under the Centre for Human-Centric AI (CHAI), automating audit analysis workflows and reducing manual effort by up to 80%.
- Architected modular document intelligence pipelines using XML preprocessing, BeautifulSoup, and pdfplumber for scalable PDF parsing, audit information processing, and automated Excel report generation.
- Implemented prompt-engineering and local inference workflows using LLaMA 3 8B through Ollama, and fine-tuned Gemma 3 1B using QLoRA and BitsAndBytes for domain-specific audit risk classification.
- Designed a RAG-based querying system using LangChain and ChromaDB for context-aware retrieval over audit reports, and developed a Gradio interface for automated PDF upload, structured report generation, and interactive querying.
- **Tech Stack:** Python, Ollama, LangChain, ChromaDB, Gradio, BeautifulSoup, pdfplumber

NIT Warangal

May 2024 – Jun 2024

Research Summer Intern

- Published IEEE conference research on illumination-aware deepfake detection using comparative CNN benchmarking across ResNet50, EfficientNet, MobileNetV2, and VGG16 architectures.
- Developed a brightness-aware evaluation pipeline using HSV-based categorization to analyze CNN robustness under varying illumination conditions in facial image manipulations.
- Achieved up to 99.9% classification accuracy using EfficientNet-based deepfake detection models under categorized illumination conditions.
- **Tech Stack:** Python, TensorFlow, Keras, OpenCV

Education

Indian Institute of Information Technology, Design and Manufacturing, Kurnool

B.Tech in Artificial Intelligence and Data Science

2022 – Present

CGPA: 9.28

Publications

- **S. B. Yangoti**, R. Dumpala and N. B. Muppalaneni, “MALLM: Malware Analysis with Large Language Models,” **International Conference on Intelligent Systems and Communication Networks (ICISCN)**, 2026. *Presented | Awaiting IEEE Publication*
- **S. B. Yangoti**, R. Dumpala, V. Punyamurthy, and R. K. Jatoth. “Illumination Effects on Deep Fake Detection: Performance Comparison of CNN Architectures.” **International Conference on Artificial Intelligence and Emerging Technology (Global AI Summit)**, 2024. *IEEE Xplore*

Projects

Triplet-Enhanced RAG for Cross-Lingual QA System GitHub

- Developed a triplet-enhanced cross-lingual RAG pipeline for multilingual question answering across Telugu, Bengali, Arabic, and Korean using the XOR-TyDi QA benchmark.
- Engineered a hybrid retrieval architecture combining BM25 retrieval, CrossEncoder reranking, REBEL-based triplet extraction, and query-aware semantic triplet ranking for structured context generation prior to answer synthesis.
- Conducted comparative ablation studies across Sparse RAG, Reranked RAG, Triplet RAG, and Triplet-Enhanced RAG architectures, achieving improvements of up to +12.61 F1 over multilingual QA baselines.

Enterprise AI Assistant with RAG and Guardrails GitHub

- Built an enterprise-style AI assistant using Retrieval-Augmented Generation (RAG) for policy-aware question answering with web-grounded contextual retrieval.
- Engineered guardrail-based action workflows with audit logging, user confirmation, and controlled task execution for secure internal automation.
- Deployed a Streamlit-based interface for interactive querying and workflow management.

MALLM: Malware Analysis with Large Language Models GitHub

- Developed a two-stage malware analysis pipeline for automated malware detection and family-wise classification using behavioral analysis of CAPEv2 sandbox execution reports.
- Fine-tuned local LLMs using LoRA-based parameter-efficient adaptation and combined LLM-generated behavioral embeddings with ML classifiers including XGBoost and Random Forest for semantic malware analysis.
- Achieved 97.85% malware detection accuracy and 97.13% malware family classification accuracy.

Achievements

- Secured **AIR 3024** in **GATE 2026 (Data Science & AI)** Credential
- Presented research paper at **ICISCN 2026** Credential
- Presented research paper at **IEEE Global AI Summit 2024** Credential
- Secured **Top 10%** in **Summer Analytics 2024**, IIT Guwahati Credential